

Breaking the 15-Minute Barrier: How Policy Changes Shape Creator Activity on YouTube

Yunxuan Hu
MSBA Department
University of Arizona
Tucson, USA
yunxuanhu@arizona.edu

Andrew Holland
MSBA Department
University of Arizona
Tucson, USA
aholland2@arizona.edu

Annie Villarreal
MIS Department
University of Arizona
Tucson, USA
drv2@arizona.edu

Abstract—This study examines how platform policy changes reshape creator behavior, focusing on YouTube’s removal of the 15-minute upload limit as a natural experiment. Using a large-scale panel dataset of 494 creators and over 80,000 videos from 2010 to 2011, we implement a difference-in-differences design to estimate the causal impact of relaxed content constraints. Contrary to conventional expectations, we find that the policy led to a significant decline in content production and engagement among general creators. However, this effect is heterogeneous: professional creators experienced increased activity and engagement following the policy change. These findings reveal a distributional consequence of platform design: policy changes that expand content affordances may disproportionately benefit resource-rich creators while disadvantaging casual participants. We propose a crowding-out mechanism, where the introduction of long-form content shifts audience attention toward professional creators, reducing visibility and incentives for others. This research contributes to the literature on platform governance and digital content production by demonstrating that platform constraints function as behavioral filters that shape not only overall activity but also inequality across users. The results have important implications for platform design, suggesting that uniform policy changes may unintentionally widen gaps between creator groups.

Keywords—Platform Governance, C2C Digital Platforms, Creator Behavior, Difference-in-Differences, Heterogeneity

I. INTRODUCTION

Digital platforms increasingly rely on policy design to shape user behavior and optimize platform performance. In consumer-to-consumer (C2C) video platforms such as YouTube, content-length constraints serve as a key design instrument that shapes both the supply of user-generated content and patterns of audience engagement. By adjusting these constraints, platforms seek to encourage participation, maintain content quality, and enhance user engagement. A prominent example is YouTube’s evolution of upload length limits. Due to technical and regulatory constraints, YouTube initially imposed short video limits. These restrictions were gradually relaxed, culminating in the removal of the 15-minute upload limit for verified users in December 2010. This policy change introduced substantial variation in content affordances across creators, creating a natural experiment to study how policy change affects user behavior.

Despite the importance of such policy changes, existing research provides limited empirical evidence on their effects in video-sharing platforms. Prior work has largely focused on theoretical models [1] or empirical settings in text-based platforms [2], showing that expanding content length can increase engagement among active users. However, it remains unclear whether these findings generalize to video platforms, where production costs, attention allocation, and content consumption dynamics differ substantially.

More importantly, existing research has largely focused on average treatment effects, implicitly assuming that platform policies influence users in a homogeneous manner. This perspective overlooks the possibility that policy changes may generate differential effects across users with varying resources, capabilities, and audience bases. In particular, extending content-length constraints may simultaneously expand opportunities for some creators while intensifying competitive pressure for others, especially in environments with limited user attention.

To address these gaps, this study examines how relaxing content-length constraints affects creator behavior and engagement on YouTube. Exploiting the removal of the 15-minute upload limit as a natural experiment, we construct a panel dataset of creators from 2010 to 2011 and implement a difference-in-differences design to estimate the causal impact of the policy change. We focus on two key outcomes: content production, measured by posting frequency and audience engagement.

Our results reveal that the effects of the policy change are not uniform. While extending content constraints is often expected to stimulate content creation and engagement, we find evidence consistent with a decline in activity among general creators, alongside positive effects for professional creators. These findings suggest that platform policy changes can have distributional consequences, redistributing attention and engagement across different types of users.

To interpret these results, we propose a mechanism based on shifts in competitive dynamics for user attention. The expansion of content affordances enables resource-rich creators to produce longer and potentially higher-quality content, which attracts disproportionate audience attention. As a result, casual creators with limited resources may experience reduced visibility and weaker incentives to participate. Accordingly, platform constraints function not only as technical limitations but also as behavioral filters that shape the competitive structure of creator ecosystems.

This study makes several contributions to the literature on digital platforms and information systems. First, it provides causal evidence on the impact of platform policy changes on user behavior in a large-scale video-sharing context. Second, it extends prior work by highlighting the importance of heterogeneity, demonstrating that platform policies can produce unequal outcomes across users. Third, it contributes to research on platform governance by showing that design decisions can unintentionally reshape the distribution of attention and participation within digital ecosystems.

II. METHODOLOGY

A. Sample Construction

We construct two complementary samples of YouTube creators using the YouTube Data API v3 to examine both general and heterogeneous policy effects. A key challenge in sample construction is ensuring representativeness while minimizing systematic selection bias.

For the baseline analysis, we construct a broad sample of creators using a randomized channel sampling strategy. Specifically, we begin with a seed generation process that queries a diverse set of publicly accessible channels, rather than focusing on pre-selected or popularity-based subsets. This approach mitigates bias toward highly active or professionally managed accounts and ensures broader coverage of casual creators. To ensure data validity, we apply a series of filtering criteria: (1) we retain only creators with observable activity prior to the policy change to establish a meaningful baseline for behavioral comparison; (2) we exclude channels with insufficient historical data to ensure adequate observations for reliable panel construction; (3) we remove channels with highly irregular or sporadic activity patterns that may undermine the stability of longitudinal analysis; and (4) we retain only channels with publicly accessible and sufficiently complete video-level metadata (e.g., timestamps, duration, and engagement measures) required for variable construction. These steps result in a balanced sample suitable for causal analysis.

To examine heterogeneity in policy effects, we construct a second sample of professional creators by identifying the top 100 YouTube channels based on subscriber rankings in 2010 and randomly selecting 30 creators from this group. Compared to the baseline sample, these creators possess greater production resources, larger audiences, and stronger incentives to adapt to platform policy changes.

For both samples, we retrieve all available videos and associated metadata, including timestamps, duration, views, likes, and comments. The combined dataset contains over 1.27 million video IDs. We restrict the analysis to the study window from January 2010 to December 2011, yielding 82,664 videos relevant to the policy change. These observations are aggregated into a creator-month panel dataset, resulting in 11,856 observations across 494 creators in the baseline sample, while the professional subsample includes 30 top creators with publicly available video-level data within the study window, used to examine heterogeneous policy effects.

B. Data Collection Pipeline

To support large-scale data collection and processing, we design a scalable data engineering pipeline that integrates data ingestion, transformation, and feature construction.

Data collection is performed using the YouTube Data API v3, through which we retrieve video-level information including timestamps, duration, views, likes, and comments. The raw dataset initially contains over 1.27 million video IDs. To efficiently handle this large volume of data, we implement a Kafka-based ingestion layer that simulates streaming data collection and ensures reliable handling of API responses. We then apply a PySpark ETL (Extract, Transform, Load) process to clean and standardize the data.

This includes JSON parsing, timestamp normalization, duration conversion, and deduplication. Within this distributed processing framework, we conduct feature engineering to construct key variables, including monthly posting counts, engagement measures, and indicators for long-form content.

Finally, the processed data are aggregated to the creator-month level, resulting in a structured panel dataset with both cross-sectional and temporal dimensions. This pipeline ensures scalability, reproducibility, and consistency in data processing, enabling subsequent causal analysis on large-scale platform data.

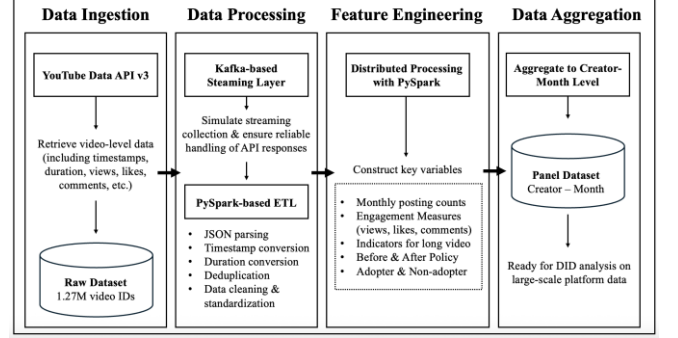


Fig. 1. Data Collection Pipeline for Creator Panel Construction

C. Empirical Framework

We construct all variables at the creator-month level. Content production is measured by the number of videos posted by creator m in month t . Audience engagement is measured by the total number of likes received by creator m in month t . To capture treatment status, we define a binary indicator based on creators' pre-policy behavior. Specifically, creators who posted video that longer than 15-minute length limit are classified as treated, as they are affected by the extension of content-length constraints. The post-policy indicator equals 1 for months after December 2010, when YouTube removed the 15-minute upload limit for verified users, and 0 otherwise.

To estimate the causal effect of the policy change, we employ a Difference-in-Differences (DID) framework. The removal of the upload limit in December 2010 provides an exogenous policy shock that differentially affects creators based on their pre-policy behavior.

We estimate the following baseline specification:

$$Y_{mt} = \beta(Treat_m \times Post_t) + \gamma_m + \delta_t + \epsilon_{mt}$$

where Y_{mt} denotes the outcome variable for creator m in month t , include measures of content production and engagement. $Treat_m$ is an indicator for treated creators, and $Post_t$ is an indicator for the post-policy period. The interaction between treatment and post-policy indicators captures the causal effect of the policy.

Two-way fixed effects models were estimated, controlling for creator-specific and time-specific unobserved heterogeneity. γ_m represents creator fixed effects, which control for time-invariant heterogeneity across creators, and δ_t represents time fixed effects, which capture common shocks across periods. The coefficient β captures the causal effect of relaxing content-length constraints on creator behavior.

We estimate the model using multiple specifications to ensure robustness. Ordinary Least Squares (OLS) models serve as the baseline, with log transformations applied to account for skewness in the outcome variables. Given that the primary dependent variables are count outcomes, we additionally estimate Poisson and Negative Binomial models to account for non-negative integer values and potential overdispersion. Standard errors are clustered at the creator level to account for serial correlation within creators over time.

D. Identification and Validation

The validity of the DID design relies on the parallel trend assumption. To assess this, we implement an event study specification that estimates dynamic treatment effects relative to the policy change. Pre-policy coefficients are statistically indistinguishable from zero, supporting the parallel trends assumption. Post-policy coefficients exhibit a clear structural shift coinciding with the policy change, reinforcing the causal interpretation of our estimates.

III. RESULTS

A. Main Effects: General Creators

We begin by examining the impact of the policy change on a broad, randomly sampled set of YouTube creators. The results reveal a negative and statistically significant effect on both content production and audience engagement.

For content production, the Difference-in-Differences model estimates indicate a substantial decline in posting activity among treated creators following the policy change. Specifically, Poisson estimates suggest a 58.48% decrease in monthly video output ($IRR \approx 0.42$, $p = 0.003$), while OLS estimates yield a consistent negative coefficient ($\beta = -0.31$, $p = 0.012$).

Similarly, audience engagement also declines significantly. Negative Binomial estimates indicate a 62.87% reduction in monthly likes ($IRR \approx 0.36$, $p = 0.0002$), with OLS estimates showing a negative but marginally significant effect ($\beta = -0.71$, $p = 0.071$). We further examine view counts as an additional measure of engagement. The results show a large and statistically significant negative effect ($\beta = -1.75$, $p = 0.0016$), suggesting that treated creators experienced a substantial decline in audience attention following the policy change.

TABLE I. DID ESTIMATION RESULTS (MAIN EFFECT)

Outcome	Model	DID Coef	p-value	IRR	% Change
Video Count	OLS	-0.310**	0.012	-	-
	Poisson	-0.879***	0.003	0.415	-58.48%
	NegBin	-0.668	-	0.513	-48.73%
Likes	OLS	-0.708*	0.071	-	-
	Poisson	0.165	0.601	1.180	+17.98%
	NegBin	-0.991***	0.0002	0.371	-62.87%
Comment	OLS	-0.538*	0.074	-	-
	Poisson	0.301	0.404	1.352	+35.19%
	NegBin	-1.011***	0.0002	0.364	-63.63%
Views	OLS	-1.752***	0.0016	-	-
	Poisson	0.129	0.666	1.138	+13.78%
	NegBin	-1.022	-	0.360	-64.01%

All models include creator and month fixed effects.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To assess the validity of the identification strategy, we implement an event study analysis. Pre-policy coefficients are statistically indistinguishable from zero, supporting the parallel trends assumption. At the time of the policy change, we observe a sharp and persistent decline in outcomes, with effects remaining negative throughout the post-policy period. This pattern reinforces the causal interpretation of the results.

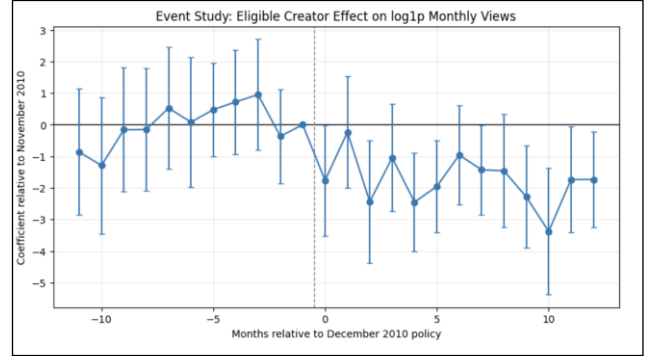


Fig. 2. Event Study of the Main Effect

TABLE II. EVENT STUDY ESTIMATES (MAIN EFFECT)

Event Time	Coefficient	Std. Error	95% CI
-11	-0.864	(1.018)	[-2.859, 1.132]
-10	-1.292	(1.100)	[-3.447, 0.864]
-9	-0.162	(1.002)	[-2.126, 1.802]
-8	-0.152	(0.990)	[-2.093, 1.789]
-7	0.525	(0.986)	[-1.407, 2.456]
-6	0.081	(1.048)	[-1.974, 2.136]
-5	0.475	(0.752)	[-1.000, 1.949]
-4	0.721	(0.844)	[-0.932, 2.375]
-3	0.951	(0.897)	[-0.806, 2.709]
-2	-0.367	(0.759)	[-1.854, 1.120]
-1	0.000	(ref)	—
0	-1.763**	(0.896)	[-3.519, -0.007]
1	-0.236	(0.903)	[-2.007, 1.534]
2	-2.442**	(0.992)	[-4.386, -0.497]
3	-1.045	(0.865)	[-2.740, 0.650]
4	-2.452***	(0.792)	[-4.004, -0.900]
5	-1.952**	(0.744)	[-3.410, -0.493]
6	-0.961	(0.804)	[-2.538, 0.615]
7	-1.427**	(0.727)	[-2.852, -0.002]
8	-1.462	(0.915)	[-3.255, 0.331]
9	-2.286**	(0.825)	[-3.904, -0.669]
10	-3.383***	(1.019)	[-5.380, -1.386]
11	-1.736**	(0.859)	[-3.419, -0.052]
12	-1.735**	(0.769)	[-3.242, -0.228]

Overall, these findings indicate that relaxing content-length constraints led to lower content production and engagement among general creators.

B. Heterogeneous Effects: Professional Creators

We next examine the heterogeneous effects of the policy change among professional creators. The Difference-in-Differences estimates indicate that, in contrast to the negative effects observed in the baseline sample, professional creators

exhibit a positive and statistically significant response in content production following the removal of the upload limit. Specifically, the estimated coefficient on the interaction term is positive and significant ($\beta = 0.286, p = 0.022$), suggesting that professional creators increased their posting activity after the policy change. Given the log-transformed dependent variable, the coefficient implies approximately a 33.10% increase in posting activity among treated professional creators relative to the control group in the post-policy period.

Turning to engagement outcomes, the results are directionally consistent but less robust. For audience engagement measured by likes, the estimated effect remains positive and statistically significant ($\beta = 0.820, p = 0.019$), indicating that professional creators not only increased content production but also attracted greater audience engagement following the policy change.

Event study results further support these findings. Similar to the baseline sample, pre-policy coefficients remain statistically indistinguishable from zero, indicating no evidence of differential pre-trends and supporting the validity of the parallel trends assumption. However, following the policy change, we observe a clear upward shift in posting activity, in contrast to the sharp decline observed among general creators.

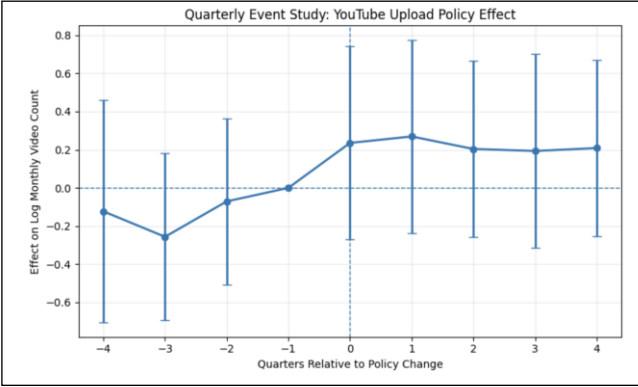


Fig. 3. Event Study of the Heterogeneous Effects

C. Conclusion

Taken together, the results across the two samples reveal a clear pattern of heterogeneous treatment effects. While the extension of content-length limit leads to significant declines in both posting activity and audience engagement among general creators, it has the opposite effect for professional creators, who exhibit increased content production and, to some extent, higher engagement following the policy change.

This divergence highlights that the impact of platform policy changes is not uniform but varies systematically across creator types. Particularly, creators with greater production resources and established audiences appear better positioned to leverage expanded content affordances, whereas casual creators may face greater challenges in adapting to longer-form content formats. As a result, the same policy intervention can generate asymmetric behavioral responses, amplifying differences in participation and performance across creators.

More broadly, these findings suggest that platform design choice function not only as technical features but also as mechanisms that shape the competitive landscape of digital

content production. By altering the conditions under which content is produced and consumed, such policies can unintentionally redistribute attention and participation in ways that favor certain groups over others.

IV. DISCUSSION

Building on the empirical findings, we turn to examining the mechanisms through which platform policy changes interact with creator characteristics to produce the observed heterogeneous effects. Although the results establish a clear divergence in behavioral responses between general and professional creators, the drivers of this divergence remain an open question. To shed light on this issue, we explore several mechanisms through which changes in platform design will interact with differences in creator resources and capabilities to generate asymmetric outcomes.

The heterogeneous effects observed in this study can be understood through shifts in competitive dynamics for user attention. Prior research suggests that digital platforms are fundamentally governed by the allocation of limited user attention, which acts as a scarce resource shaping content visibility and creator success [3]. The relaxation of content-length constraints enables professional creators to produce longer and potentially higher-quality content, thereby capturing a larger share of audience attention. As attention becomes increasingly concentrated on such content, casual creators may experience reduced visibility, even in the absence of direct changes to their own behavior. This mechanism aligns with emerging evidence that platform design features can significantly influence how attention is distributed across participants [4], suggesting that policy changes may indirectly reshape competitive outcomes through attention reallocation.

A complementary explanation is based on creators' trade-offs between expected benefit and effort investment. The relaxation of content-length constraints changes the payoff structure of content production by increasing the potential returns to longer-form content, particularly for creators with established audiences. For professional creators, the expected gains from producing longer and more engaging content, such as higher visibility, stronger audience retention, and greater monetization opportunities, may outweigh the additional effort required. In contrast, casual creators may face a different cost-benefit calculus. Producing longer-form content often requires greater time investment, higher technical skills, and production resources, while the expected returns in terms of visibility and engagement remain uncertain or relatively low. As a result, the marginal benefit of increasing effort may not justify the additional cost, leading to reduced participation or lower posting activity. This interpretation is consistent with prior research showing that individuals' participation decisions in digital environments are shaped by perceived returns relative to effort and resource investment.

This asymmetry in cost-benefit evaluation perspective is consistent with prior research on user heterogeneity in digital environments, which shows that resource-rich participants are better positioned to respond to changes in platform incentives and technological affordances [5]. As a result, policies intended to expand creative possibilities may inadvertently raise the effective barrier to participation for less-resourced users, leading to reduced activity among casual creators.

More broadly, our findings contribute to the literature on platform governance and digital inequality by demonstrating that seemingly neutral design choices can have uneven behavioral consequences. Platform design, such as limits on content length, do not merely regulate technical functionality but also shape the structure of participation and competition within digital ecosystems [6]. By altering the relative returns to different types of content production, policy changes can unintentionally amplify disparities between creator groups.

Taken together, our results highlight the importance of considering heterogeneity in user capabilities when designing platform policies, as uniform changes may generate unequal impacts across participants.

V. CONCLUSION

This study investigates how platform policy changes shape creator behavior by leveraging YouTube’s removal of content-length constraints as a natural experiment. Using a Difference-in-Differences analysis framework, we document systematic heterogeneity in responses across creator types. While casual creators experience declines in content production and engagement, professional creators increase their activity following the policy change, highlighting a clear divergence in behavioral responses.

These findings suggest that platform policy changes do not affect all participants uniformly. Instead, they reshape the incentive environment in ways that interact with differences in creator capabilities, leading to asymmetric outcomes across users. By altering both the potential returns to content production and the conditions under which attention is allocated, such policies can differentially advantage creators with greater resources while discouraging participation among others. In this context, platform design functions not only as a technical constraint but also as a mechanism that structures competition and participation within digital ecosystems.

Our results contribute to the literature on platform governance by demonstrating that seemingly neutral design choices can have uneven and unintended consequences.

From a managerial perspective, these findings highlight the importance of considering user heterogeneity in the platform when implementing policy changes, as expanding creative affordances may come at the cost of reduced inclusivity.

This study has several limitations. The analysis focuses on a single policy change within one platform, and the proposed mechanisms are inferred rather than directly tested. Future research could examine other platform settings, explore alternative policy interventions, and develop more direct measures of creators’ capabilities and strategic responses.

REFERENCES

- [1] Gu, Z., & Zhao, X. (2024). Content length limit: how does it matter for a consumer-to-consumer media platform?. *Information Systems Research*, 35(4), 1785-1801.
- [2] Wang, S., & Greenwood, B. N. (2025). Too long, didn’t read? The impact of length limits on microblogging engagement. *Management Science*, 71(11), 9437-9461.
- [3] Davenport, T. H., & Beck, J. C. (2001). The attention economy. *Ubiquity*, 2001(May), 1-es.
- [4] Monge Roffarello, A., De Russis, L., & Lukoff, K. (2025). The digital attention heuristics: Supporting the user’s attention by design. *ACM Transactions on Computer-Human Interaction*, 32(4), 1-41.
- [5] Rochet, J. C., & Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4), 990-1029.
- [6] Tucker, C., & Zhang, J. (2011). How does popularity information affect choices? A field experiment. *Management Science*, 57(5), 828-842.

INDIVIDUAL CONTRIBUTION

Yunxuan Hu: Literature Review and Background Research; Data Collection Pipeline Design; Conduct Heterogeneous Effects Estimation; Final Report Draft

Andrew Holland: Analysis Pipeline Developer; Poisson, Negative-Binomial, and Parallel Trends Analysis; Data API Calling and Data Hosting

Annie Villarreal: Project Lead; Problem Formulation and Hypothesis Development; Proposal and Progress Report Writing; Final Report Editing; Presentation Design and Slide Deck Construction